

AI-Powered Document Intelligence and Semantic Search System using RAG

Product Requirements Document (PRD)

AI-Powered Document Intelligence and Semantic Search System using RAG

1. Project Overview

Project Title:

AI-Powered Document Intelligence and Semantic Search System using Retrieval-Augmented Generation (RAG)

Project Type:

Final Year Placement-Oriented AI Project

Project Category:

- Artificial Intelligence
- Natural Language Processing
- Retrieval-Augmented Generation (RAG)
- Semantic Search
- Document Intelligence

2. Executive Summary

The AI-Powered Document Intelligence and Semantic Search System using RAG is an intelligent enterprise-oriented platform designed to process, classify, retrieve, and analyze unstructured documents using Artificial Intelligence and Natural Language Processing techniques.

Traditional document search systems rely on keyword-based retrieval methods that fail to understand semantic meaning and contextual relevance. This project introduces a semantic document intelligence platform capable of understanding user intent and retrieving contextually relevant information from uploaded documents through vector embeddings and Retrieval-Augmented Generation (RAG).

The system enables users to upload documents such as PDFs, research papers, resumes, and reports, automatically classify them into categories, perform semantic search, and ask context-aware questions from the uploaded content.

3. Problem Statement

Organizations and institutions generate large amounts of unstructured documents daily. Existing document management systems mainly use keyword-based retrieval techniques that:

- Fail to understand semantic meaning
- Produce inaccurate search results
- Require manual searching
- Increase document retrieval time

There is a need for an intelligent system capable of:

- Understanding document context
- Performing semantic retrieval
- Generating context-aware responses
- Reducing manual document analysis effort

4. Proposed Solution

The proposed solution is an AI-powered document intelligence platform that combines:

- Document processing
- Semantic embeddings
- Vector databases
- Retrieval-Augmented Generation (RAG)

The system:

1. Accepts document uploads
2. Extracts and processes text
3. Generates semantic embeddings
4. Stores embedding in a vector database
5. Retrieves relevant document chunks
6. Generates context-aware responses using RAG

5. Objectives

Primary Objectives:

- Build an intelligent document processing system
- Implement semantic search using vector embeddings
- Integrate Retrieval-Augmented Generation (RAG)
- Enable context-aware document querying
- Improve retrieval relevance compared to keyword search

Secondary Objectives:

- Build scalable modular architecture
- Support multiple document formats
- Create a placement-oriented production-style project
- Demonstrate practical AI system implementation

6. Functional Requirements

- Document Upload
- Text Extraction
- Document Chunking
- Embedding Generation
- Semantic Search
- Document Classification
- RAG-Based Querying

7. Technology Stack

Frontend: React + Tailwind

Backend: FastAPI

Embeddings: Sentence Transformers

Vector Database: FAISS or ChromaDb

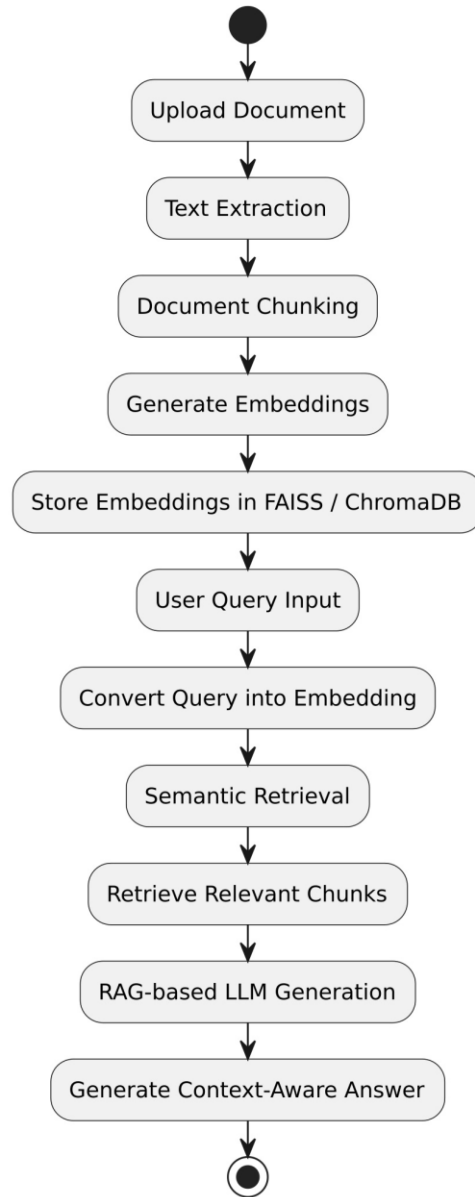
RAG Framework: LangChain

LLM: Gemini/OpenAI/Groq

PDF Parsing: pdfplumber/PyPDF

8. End-to-End Workflow

AI-Powered Document Intelligence and Semantic Search System using RAG



9. Project Structure for the Final MVP

```
document-intelligence-rag/  
├── frontend/  
│   ├── src/  
│   │   ├── components/  
│   │   │   ├── Upload.jsx  
│   │   │   ├── Search.jsx  
│   │   │   ├── Chat.jsx  
│   │   │   └── Navbar.jsx  
│   │   ├── pages/  
│   │   │   └── Home.jsx  
│   │   ├── services/  
│   │   │   └── api.js  
│   │   ├── App.jsx  
│   │   └── main.jsx  
│   ├── package.json  
│   └── vite.config.js  
├── backend/  
│   ├── services/  
│   │   ├── document_processor.py  
│   │   ├── chunking_service.py  
│   │   ├── embedding_service.py  
│   │   ├── vector_service.py  
│   │   ├── retrieval_service.py  
│   │   └── rag_service.py  
│   ├── routes/  
│   └── main.py  
├── vector_store/  
├── data/  
└── README.md
```

10. Edge Cases

- Wrong classification
- OCR/text extraction failure
- Duplicate documents
- Long document context loss
- Poor retrieval quality
- Hallucinated responses
- Unsupported files
- Empty documents

11. Future Enhancements

- OCR integration
- Multi-language support
- Conversational memory
- Multi-document querying
- Analytics dashboard
- Authentication system
- Hybrid retrieval
- Cloud deployment

12. Conclusion

The AI-Powered Document Intelligence and Semantic Search System using RAG is a modern AI-driven platform designed to improve document understanding, retrieval, and contextual querying through semantic search and Retrieval-Augmented Generation.